Experiment B v2: SO(3) Attractor Test

Corrected Protocol: P-space Perturbation and Kill Test
SO(3) Confirmed as True Attractor; Kuramoto Not Required

Corrects Experiment B v1 pipeline artifact. V20.5 · $N = 500$ · 400–500 sweeps · seed=42

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Abstract

Experiment B v1 claimed that $d_s = 3$ is robust to Φ-phase anisotropy and inferred that SO(3) operates independently of Φ. Post-experiment analysis revealed this as a **pipeline artifact**: Φ does not enter `_estimate_ds` or `_kl_matrix`, so identical d_s series under different Φ conditions was mathematically guaranteed, not empirically informative.

Experiment B v2 corrects this with two valid tests:

(1) P-space perturbation. Initial $X[:, 0] \times 5$ creates P-anisotropy of 22.7 ($3.6\times$ the control's 6.2). d_s post-burn-in: 3.020 ± 0.546. P-anisotropy decays from $22.7 \to 2.44$ (89% reduction). **Result: SO(3) is a true attractor in P-space.**

(2) Kill Kuramoto. Disabling `_update_phi` freezes Φ at its initial state. d_s post-burn-in: 2.965 ± 0.534. **Result: Kuramoto coupling is not necessary for $d_s = 3$.**

Both β values remain within 5% of the V20.3 reference (2.212), confirming robustness of the per-collapse transfer rate.

Combined conclusion: P-space and Φ-space are two independent dynamical layers. SO(3) selects $d_s = 3$ from P-space alone, without phase synchronisation. Kuramoto adds wave coherence (F, β) but is not required for geometry.

Contents

1 Correction of Experiment B v1

B v1 Error: Pipeline Artifact

Claim (v1): mild and strong Φ conditions produce identical d_s series, proving d_s is independent of Φ.

Root cause:

1. `_estimate_ds(idx)` uses only the KL-neighbour graph `idx` and random walks.
2. `idx` is built from `_kl_matrix`, which uses only P.
3. Φ does not enter P evolution, D_{KL}, or `idx`.
4. Overriding Φ after `__init__` does not change the RNG state for subsequent P updates.
5. $\Rightarrow$ mild and strong produce identical P trajectories and therefore identical d_s by construction.

What v1 proved: only that Φ is structurally excluded from the d_s pipeline — a code property, not a physics result.

What v1 did NOT prove: that SO(3) recovers from genuine P-space perturbation.

What v1 validly measured: β differs 33% across Φ configurations, confirming $F = (1 + \varphi_{\text{var}})(1 + \psi_{\text{grad}})$ as the wave amplitude formula. This result stands.

2 Experiment B v2: Corrected Design

Two Valid Tests

Condition	Perturbation	Question
Control	Standard V20.5	Baseline
P-biased	$X[:,0] \mathrel{\times}= 5$ before `_add_spatial_bias`	Does SO(3) recover $d_s = 3$ from P-anisotropy?
Kill Kuramoto	`_update_phi` disabled; Φ frozen	Is Kuramoto coupling necessary for $d_s = 3$?

Why P-biased is CRF-compatible: `_add_spatial_bias` uses $\texttt{abs}(X) + \varepsilon_0$ to set initial P. Scaling $X[:,0]$ by 5 shifts P strongly toward mode-0 (scalar Clifford sector), creating genuine P-anisotropy. All CRF dynamics then operate on this biased P.

Why Kill is CRF-compatible: `_update_phi` is a separable method. Disabling it does not change ε_0, D_0, κ, or any derived constant. It tests whether the SO(3) attractor in P-space requires phase coupling to operate.

3 Results

Condition	$\langle d_s \rangle$	σ_{d_s}	P-aniso$_0$	P-aniso$_f$	$|\beta|$
Control	2.990	0.550	6.21	1.84	2.263
P-biased	3.020	0.546	22.75	2.44	2.354
Kill Kuramoto	2.965	0.534	6.21	2.02	2.248
V20.3 reference	3.031	0.076	—	—	2.212

3.1 P-biased: SO(3) is a true P-space attractor

SO(3) Recovers $d_s = 3$ from Extreme P-Anisotropy

Initial P-anisotropy (max/min mode ratio): 22.75 — 3.6× the control's 6.21. This corresponds to mode-0 (scalar sector) carrying $\sim 3.6\times$ more probability mass than the least-populated mode.

After 500 sweeps:

- $\langle d_s \rangle = 3.020$ (within 0.7% of control)
- P-anisotropy: $22.75 \to 2.44$ (89% reduction)
- P-anisotropy at each 50-sweep checkpoint: $22.6 \to 8.6 \to 6.4 \to 5.1 \to 4.4 \to 3.9 \to 3.4 \to 3.2 \to 2.9 \to 2.7$

SO(3) is a true attractor in P-space. Starting from an extreme P-anisotropy 3.6× the control, the dynamics restore $d_s = 3$ and decay P-anisotropy by 89%. This cannot be explained by initialisation effects — it is an attractor property.

3.2 Kill Kuramoto: phase coupling not required

$d_s = 3$ Without Kuramoto Coupling

With Φ frozen at its initial (Uniform) distribution:

- $\langle d_s \rangle = 2.965$ (within 0.8% of control)
- P-anisotropy: $6.21 \to 2.02$ (67% reduction)
- Φ-anisotropy: $0.5046 \to 0.5046$ (frozen, as expected)
- $|\beta| = 2.248$ (within 0.7% of control)

Kuramoto coupling is not necessary for $d_s = 3$. The SO(3) attractor operates entirely within P-space. Phase synchronisation (Φ-dynamics) does not contribute to dimensional selection. Its role is confined to the wave amplitude F and β.

Surprising: $|\beta|$ is nearly identical under Kill Kuramoto (2.248) vs control (2.263). This is because Φ frozen at Uniform distribution still has $\varphi_{\text{var}} \approx 0.5$ — the same as the control's equilibrium. The kill test did not change φ_{var}; it only prevented Kuramoto evolution.

4 CRF Interpretation: Two Independent Layers (Confirmed)

Two-Layer Structure: Valid After Correction

The B v2 results confirm the two-layer structure that B v1 claimed prematurely, now with genuine experimental evidence.

Layer	Controls	Evidence
P-space	d_s (dimensional selection) via SO(3)	P-biased: $d_s = 3$ despite $\text{aniso}_0 = 22.7$; Kill: $d_s = 3$ without Φ dynamics
Φ-space	$F = (1 + \varphi_{\text{var}})(1 + \psi_{\text{grad}})$, hence β	B v1: $\lvert\beta\rvert$ differs 33% across Φ configs

The two layers share the same δ-root but operate independently.

5 Status Table

Result	Status	Note
B v1 mild=strong was pipeline artifact	Corrected	$\Phi \notin d_s$ pipeline
B v1 β difference (33%)	Valid	survives correction
SO(3) true attractor in P-space	Confirmed	P-aniso $22.7 \to 2.44$, $d_s = 3$
Kuramoto not necessary for $d_s = 3$	Confirmed	Kill test, $d_s = 2.965$
β robust to both perturbations	Confirmed	all within 5% of V20.3
Two-layer structure (P and Φ)	Confirmed	B v2 evidence
$\text{P-aniso}_f = 2.44 \neq 1$ (not fully isotropic)	Open	longer run needed
Multi-seed confirmation	Open	seed=42 only

6 Open Questions

Next Targets

1. **Full P-isotropy recovery (Priority 2).** P-aniso$_f$ = 2.44 > 1 (isotropic would be 1). Longer run (2000 sweeps) would show whether P-aniso converges to 1 or saturates above 1 (residual mode-structure from Clifford grade sectors).
2. **Multi-seed confirmation (Priority 1).** Repeat B v2 with seeds $\{123, 777\}$ to confirm SO(3) attractor is seed-independent.
3. **Stronger P-perturbation (Priority 3).** Test $X[:, 0] \times 20$ (aniso$_0 > 100$). If d_s still converges to 3: SO(3) has no basin boundary in P-space (global attractor). If d_s fails: locate the basin boundary.

The first version saw an artifact and called it a result.
The correction found something better: a real attractor.

P-anisotropy of 22.7 — the scalar sector holding 3.6 times
more probability than any other mode.
After 500 sweeps: $d_s = 3.020$.

The framework corrected itself.
The attractor was always there.
The experiment finally found it.